MICROBIOLOGY · MCQ SET

Staphylococcus saprophyticus & Enterococcus

LG 2 · Dr Samira Fattah · GUE BLOCK · PRACTICE MCQs

20 questions, mixed difficulty, 9 case-based. Attempt every question before checking the answer key on the last page.

4	10	6
Easy	Medium	Hard

1. EASY

What is the catalase reaction of S. saprophyticus, and of Enterococcus?

A. Both positive
B. Both negative
C. S. saprophyticus positive, Enterococcus negative
D. S. saprophyticus negative, Enterococcus positive
E. Both variable

2. EASY

Which organism is classically associated with uncomplicated cystitis in young, sexually active women?

A. Enterococcus faecalis
B. Enterococcus faecium
C. Staphylococcus saprophyticus
D. Staphylococcus aureus
E. Staphylococcus epidermidis

3. EASY

What is the main natural reservoir of Enterococcus?

A. Skin
B. Respiratory tract
C. Gastrointestinal tract
D. Oral cavity
E. Genital tract only

4. EASY

Which antibiotic susceptibility test is used as a presumptive feature to identify S. saprophyticus among coagulase-negative staphylococci?

A. Oxacillin
B. Novobiocin
C. Vancomycin
D. Bacitracin
E. Optochin

5. MEDIUM

A coagulase-negative, catalase-positive Gram-positive coccus in clusters is isolated from a urine culture and found to be novobiocin resistant. What is the most likely organism?

A. S. aureus
B. S. epidermidis
C. S. saprophyticus
D. Enterococcus faecalis
E. Streptococcus agalactiae

6. MEDIUM CASE

A young woman has classic cystitis symptoms. Urinalysis shows pyuria but a negative nitrite test.

What is the most appropriate interpretation?

A. Infection is excluded, since nitrite is negative
B. Nitrite negativity is consistent with S. saprophyticus, which does not typically reduce nitrate
C. The sample must be repeated due to lab error
D. Nitrite negativity proves this is Enterococcus
E. Nitrite negativity suggests a viral cause

7. MEDIUM

Which feature is shared by both S. saprophyticus and Enterococcus on routine Gram stain and growth characteristics?

A. Both grow in clusters and are catalase positive
B. Both are Gram-positive cocci
C. Both are strongly hemolytic on blood agar
D. Both are motile
E. Both are oxidase positive

8. MEDIUM CASE

A patient has a blood culture growing Enterococcus faecalis, and the clinical team considers starting a cephalosporin while awaiting full susceptibilities.

Why would a cephalosporin not be an appropriate empirical choice here, even before susceptibility results return?

A. Cephalosporins are never active against Gram-positive organisms

B. Enterococcus has intrinsic resistance to cephalosporins

C. Cephalosporins are contraindicated in all UTIs

D. Cephalosporins only treat anaerobes

E. The patient is allergic by default

9. MEDIUM

What is the recognized risk factor profile that increases risk of healthcare-associated Enterococcus infection?

A. Young age and good health

B. Urinary catheters, prolonged admission, antibiotic exposure, impaired host defenses

C. Outpatient status only

D. Lack of any hospital contact

E. Vegetarian diet

10. MEDIUM CASE

A patient with persistent Enterococcus faecalis bacteremia and no clear removable source is being evaluated for a possible cardiac source.

What is the most likely valvular/cardiac complication to consider in this scenario?

A. Myocarditis

B. Infective endocarditis

C. Pericarditis

D. Rheumatic fever

E. Cardiac tamponade

11. MEDIUM

A urine culture grows Enterococcus faecalis susceptible to ampicillin and nitrofurantoin in a patient with uncomplicated cystitis. What is a reasonable treatment approach?

A. IV vancomycin only

B. Amoxicillin or nitrofurantoin, since the isolate is susceptible and this is localized cystitis

C. No treatment, since Enterococcus is never pathogenic in urine

D. A carbapenem empirically

E. High-dose gentamicin monotherapy

12. MEDIUM CASE

A 70-year-old hospitalized man with a long-term urinary catheter has a routine urine culture growing Enterococcus faecium. He is afebrile with no urinary symptoms, confusion or systemic signs.

What is the most appropriate next step?

A. Start IV vancomycin immediately

B. No antibiotics; evaluate catheter need and consider removal/replacement

C. Start nitrofurantoin for 7 days

D. Repeat culture only if he develops hematuria

E. Start combination ampicillin plus gentamicin

13. HARD

Which combination of lab findings would specifically argue AGAINST a diagnosis of S. saprophyticus in a young woman's urine culture?

A. Catalase positive, coagulase negative, novobiocin resistant

B. Catalase positive, coagulase negative, novobiocin sensitive

C. Gram-positive cocci in clusters, nonhemolytic

D. Isolated from a young, sexually active woman

E. Urine culture showing pyuria

14. HARD CASE

A hospital infection control team identifies an Enterococcus faecium isolate with high-level vancomycin resistance (VRE).

What is the molecular mechanism underlying this resistance?

A. Efflux pump actively exporting vancomycin

B. Beta-lactamase production that inactivates vancomycin

C. Replacement of the D-Ala-D-Ala peptidoglycan precursor with D-Ala-D-Lac, which vancomycin cannot bind

D. Loss of peptidoglycan synthesis altogether

E. Mutation of ribosomal binding sites

15. HARD CASE

A patient with Enterococcus faecium bacteremia fails to clear the organism despite what should be adequate ampicillin dosing, and AST later shows ampicillin resistance.

Besides intrinsic cephalosporin resistance, what additional mechanism could explain persistent bacteremia despite adequate ampicillin dosing in this isolate?

A. Altered penicillin-binding proteins causing ampicillin resistance

B. The patient is simply non-adherent to oral therapy

C. Enterococcus cannot cause bacteremia

D. The isolate must be a contaminant

E. Ampicillin never covers Gram-positive organisms

16. HARD

Why can low-level aminoglycoside resistance in Enterococcus still sometimes be overcome in combination therapy, whereas high-level acquired resistance cannot?

A. Low-level resistance has no clinical relevance at all

B. Combination with a cell-wall-active agent (e.g. ampicillin) permits synergistic killing that high-level acquired resistance specifically abolishes

C. High-level resistance only affects Gram-negative organisms

D. Aminoglycosides are never used for Enterococcus regardless of resistance level

E. Low-level resistance converts automatically to vancomycin resistance

17. HARD CASE

An Enterococcus isolate persists on an indwelling urinary catheter despite a course of an active antibiotic, with recurrent positive cultures from the same device.

What virulence factors most plausibly explain the isolate's persistence on the device despite antibiotic exposure?

A. Hemolysin production alone

B. Adhesins and biofilm formation on the prosthetic/catheter surface

C. Spore formation

D. Capsule-mediated motility

E. Flagellar motility

18. HARD

Which statement about treating Enterococcus urinary isolates is CORRECT?

A. All Enterococcus bacteriuria must be treated regardless of symptoms

B. Treatment should distinguish symptomatic cystitis, systemic UTI and colonization before deciding on therapy

C. Nitrofurantoin is never appropriate for Enterococcus

D. Cephalosporins are first-line for Enterococcus UTI

E. Asymptomatic bacteriuria should always prompt vancomycin

19. HARD CASE

A urine isolate is Gram-positive cocci in clusters, catalase positive, coagulase negative, and resistant to novobiocin, from a 24-year-old sexually active woman with dysuria.

Which organism best fits this isolate, and which single test result above is most specific for ruling it in?

A. S. aureus; coagulase positivity

B. S. saprophyticus; novobiocin resistance

C. Enterococcus faecalis; catalase positivity

D. S. epidermidis; novobiocin resistance

E. Streptococcus pyogenes; catalase negativity

20. HARD

A microbiology report describes a Gram-positive coccus in pairs/short chains, catalase negative, non-hemolytic on blood agar, from a patient's blood culture with suspected endocarditis. Which organism and species is most likely, and why does this matter for prognosis?

A. S. saprophyticus; benign, no endocarditis risk

B. Enterococcus faecalis; a recognized cause of infective endocarditis

C. S. epidermidis; always a contaminant

D. Streptococcus pneumoniae; unrelated to endocarditis

E. Enterococcus faecium; never causes endocarditis

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	D	S. saprophyticus is catalase positive; Enterococcus is catalase negative in routine identification.
2	C	S. saprophyticus is classically associated with cystitis in young, sexually active women.
3	C	Enterococcus is part of normal human intestinal flora, which is its main reservoir.
4	B	Novobiocin resistance is the key presumptive feature distinguishing S. saprophyticus from other coagulase-negative staphylococci.
5	C	Catalase-positive, coagulase-negative, clustered Gram-positive cocci with novobiocin resistance is the classic S. saprophyticus profile.
6	B	S. saprophyticus often gives a negative urine nitrite because it does not typically reduce nitrate, unlike the Enterobacterales — infection is not excluded by a negative nitrite.
7	B	Both organisms are Gram-positive cocci; their growth arrangement (clusters vs pairs/chains), catalase reaction, and hemolysis pattern differ.
8	B	Enterococcus has intrinsic resistance that makes cephalosporin monotherapy ineffective, regardless of in vitro susceptibility testing against other classes.
9	B	Urinary catheters, prolonged hospital admission, antibiotic exposure, and impaired host defenses are the stated risk factors for healthcare-associated Enterococcus infection.
10	B	Enterococcus, especially E. faecalis, is a recognized cause of infective endocarditis, making this the key complication to evaluate for in persistent bacteremia.
11	B	Susceptible Enterococcus causing localized cystitis may be treated with amoxicillin or nitrofurantoin in selected cases.
12	B	Asymptomatic bacteriuria in an afebrile, asymptomatic catheterized patient should prompt catheter assessment rather than antibiotics outside guideline exceptions.
13	B	Novobiocin sensitivity argues against S. saprophyticus, since resistance is the defining presumptive feature; a sensitive result points to another coagulase-negative staphylococcus.
14	C	VanA/VanB mechanisms replace the D-Ala-D-Ala peptidoglycan precursor with D-Ala-D-Lac, which vancomycin cannot bind, abolishing inhibition of cell-wall synthesis.
15	A	Altered penicillin-binding proteins can cause ampicillin resistance in Enterococcus, a separate mechanism from the aminoglycoside and vancomycin resistance pathways.
16	B	Combining a cell-wall-active agent with an aminoglycoside permits synergistic killing in strains with only low-level aminoglycoside resistance; acquired high-level resistance specifically abolishes this synergy.
17	B	Adhesins and biofilm formation support Enterococcus persistence on catheters and prosthetic material despite antibiotic exposure.
18	B	Treatment should first distinguish symptomatic cystitis, systemic UTI, and colonization, since this determines whether and how to treat.
19	B	This profile (clustered Gram-positive cocci, catalase positive, coagulase negative, novobiocin resistant) in a young sexually active woman is S. saprophyticus, with novobiocin resistance the most specific confirming feature.
20	B	Catalase-negative Gram-positive cocci in pairs/chains, non-hemolytic, from blood with suspected endocarditis fits Enterococcus (commonly E. faecalis), a recognized cause of infective endocarditis.